703330 VU Advanced Topics in Perception, Interaction and Robotics: Machine Learning for Interactive Systems

Introduction

Justus Piater

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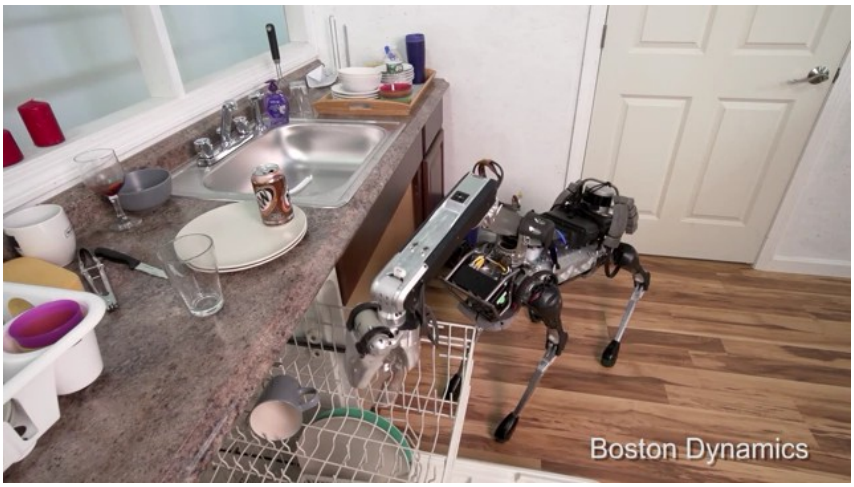
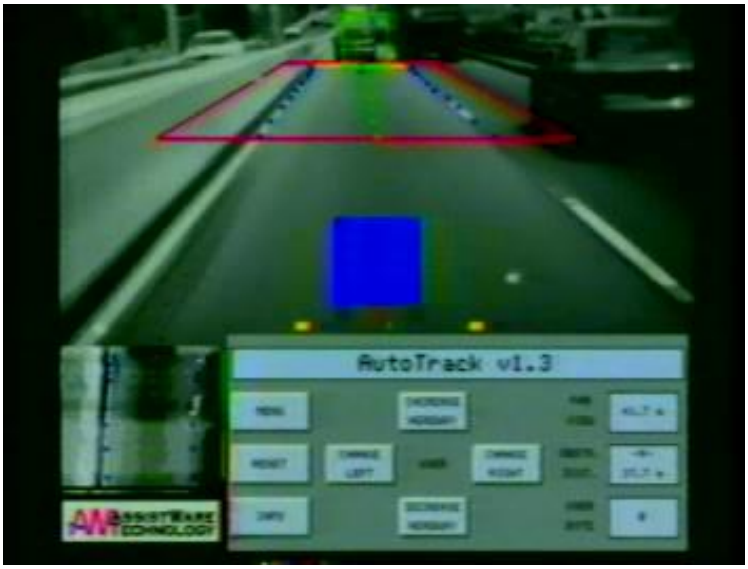
1 Introduction

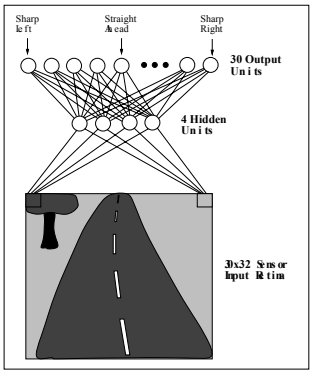
How to learn such skills? [Slide 1]



[Boston Dynamics; excerpt from Youtube]

ALVINN [Slide 2]





[Sukthankar et al. 1993]

Quadruped Walking [Slide 3]

Before Learning

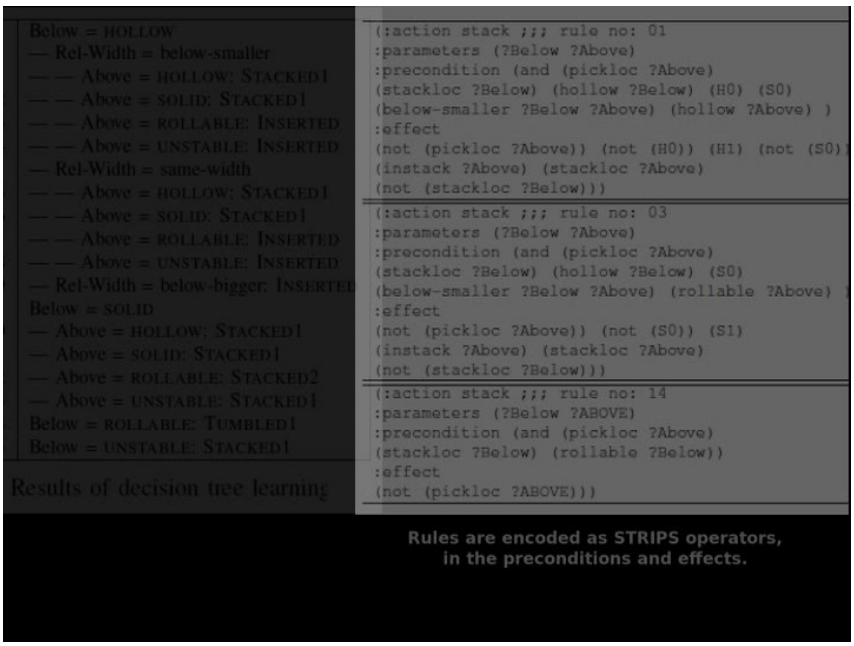
After Learning

[Kohl and Stone 2004]

Motor Skill Learning [Slide 4]

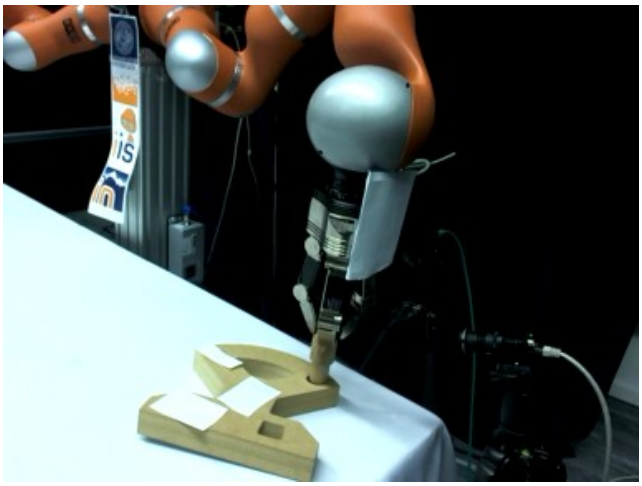
[Kober and Peters 2008]

Learning to Play With Building Blocks [Slide 5]



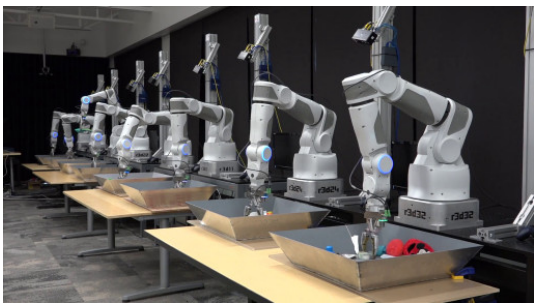
[Ugur and Piater 2015]

Learning to Assess and Improve Task Outcomes [Slide 6]



[Erkent et al. 2017]

Learning hand-eye coordination for robotic grasping with deep learning and large-scale data collection [Slide 7]

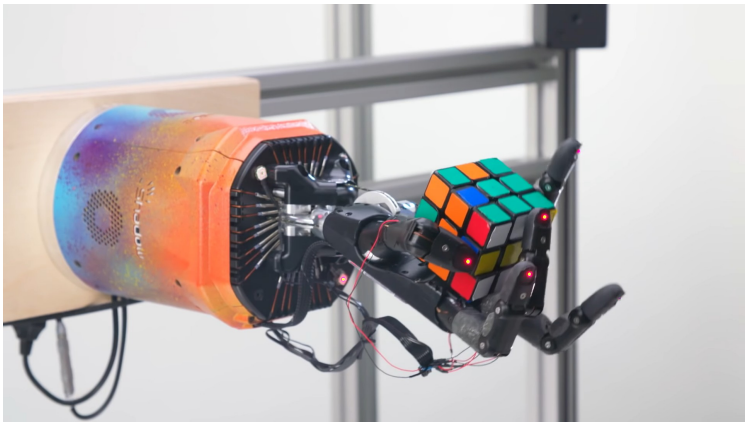


- end-to-end 4-DoF (top-down) grasp learning from monocular images
- generates reactive grasping policies
- CNN predicts the probability that task-space gripper motion will result in a successful grasp
- trained on 800 000 real grasps
- success rate 80–90%

Video (training); Video (testing) [Levine et al. 2018]

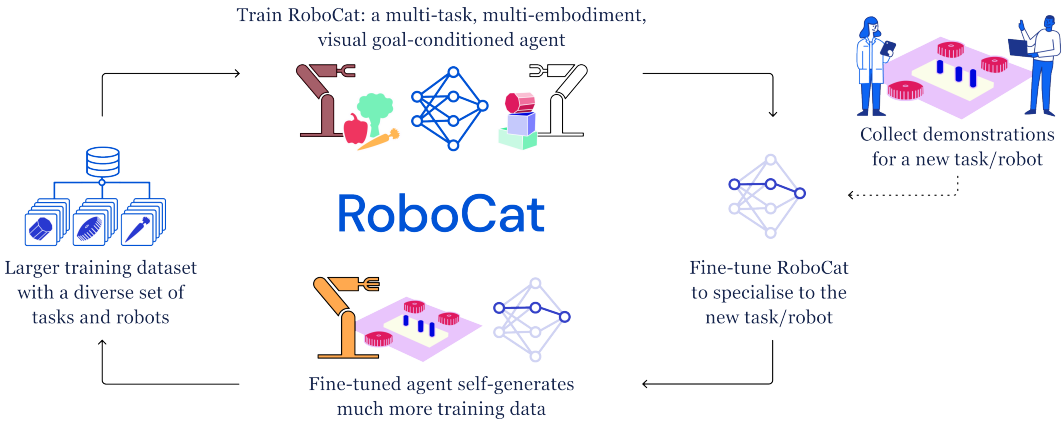
Dexterous Manipulation [Slide 8]

Lots of training in simulation with *domain randomization*



[OpenAI 2019]

RoboCat (Google DeepMind 2023-06-20) [Slide 9]



[Bousmalis et al. 2023]

Core: a *transformer* that chooses actions autoregressively:

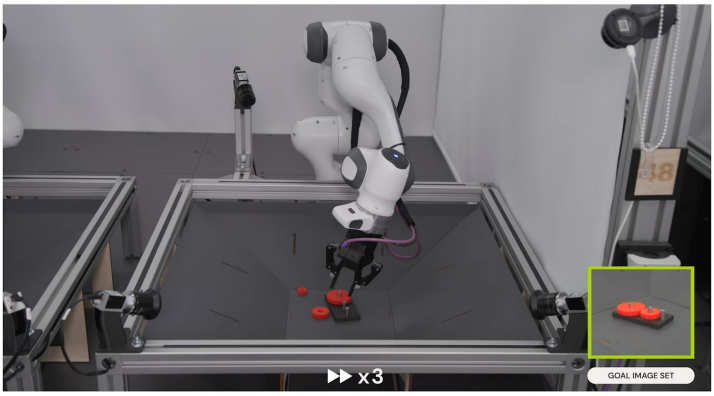
$$\pi(a_t \mid x_t, I_t, g_t) = p(a_t \mid x_{<t}, I_{<t}, g_{<t})$$

$$a_t = \text{action at time } t$$

$$x, I = \text{current state: robot configuration and camera image}$$

$$g = \text{goal state: camera image}$$

RoboCat in Action [Slide 10]



[YouTube (excerpts)]

2 Course Organization

Course Organization [Slide 11]

All information is in OLAT.

Course Content (tentative) [Slide 12]

Some Basics of Machine Learning Revisited

- ML and MAP parameter estimation; Bayesian prediction
- generative vs. discriminative models
- Neural Network Basics

Reinforcement Learning

- Deep RL
- Policy-Search RL
- Actor-Critic RL

Some Hot Topics (student presentations?)

- Transformers in Robotics
- Disentangled Representation Learning
- Causality Theory
- Inductive Logic Programming

3 References

References [Slide 13]

Bousmalis, Konstantinos, Giulia Vezzani, Dushyant Rao, Coline Devin, Alex Lee, Maria Bauza, Todor Davchev, Yuxiang Zhou, Agrim Gupta, Akhil Raju, Antoine Laurens, Claudio Fantacci, Valentin Dalibard, Martina Zambelli, Murilo Martins, Rugile Pevceviciute, Michiel Blokzijl, Misha Denil, Nathan Batchelor, Thomas Lampe, Emilio Parisotto, Konrad Żołna, Scott Reed, Sergio Colmenarejo, Jon Scholz, Abbas Abdolmaleki, Oliver Groth, Jean-Baptiste Regli, Oleg Sushkov, Tom Rothörl, José Chen, Yusuf Aytar, Dave Barker, Joy Ortiz, Martin Riedmiller, Jost Springenberg, Raia Hadsell, Francesco Nori, and Nicolas Heess (June 2023). *RoboCat: A Self-Improving Foundation Agent for Robotic Manipulation.* arXiv:2306.11706. DOI: 10.48550/arXiv.2306.11706. URL: https://arxiv.org/pdf/2306.11706.pdf.

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Ugur, Emre and Justus Piater (May 2015). "Bottom-Up Learning of Object Categories, Action Effects and Logical Rules: From Continuous Manipulative Exploration to Symbolic Planning." In: *International Conference on Robotics and Automation.* IEEE, pp. 2627–2633. DOI: 10.1109/ICRA.2015.7139553. URL: https://iis.uibk.ac.at/public/papers/Ugur-2015-ICRA.pdf.